

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a2b3f185d0eaf72a2.jsonl`  
- SHA-256: `9a04cf24e9295550741c5b3bdaa4f4f7fb15c1c104565cba6313b134bd867cd4`  
- Source modified: `2026-06-06T17:37:54+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:36:35.930Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"A lightweight six-layer CNN over Mel spectrograms classifies racket-bounce sound events with very high accuracy (spin F1=0.96, surface F1=0.97), strongly outperforming classical methods (SVM ~0.93-0.94, GMM 0.06-0.17), showing simple locally-runnable models suffice for audio event classification once peaks are isolated."

****Source:**** <https://arxiv.org/html/2409.11760v1> (primary)
****Supporting quote:**** "We utilized a six-layer CNN for bounce sound classification... 64 Mel bands and an FFT window length of 256 samples... Spin | 0.98 | 0.95 | 0.96 ... Surface | 0.97 | 0.97 | 0.97"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:36:38.817Z)

I'll verify this claim by examining the source and searching for contradicting evidence. Let me start by loading the web tools.

3. assistant (2026-06-06T17:36:46.368Z)

I'll fetch the primary source first to check whether the quote and claim are accurate, and what domain this paper actually covers.

4. user (2026-06-06T17:36:54.230Z)

Paper Summary

**** (1) Sport/Domain:**** Table tennis (ping pong)

**** (2) Bounce Sound Classification Task:****

- ****Spin classification****: Determines whether spin was applied to the ball (categories: backspin, topspin, or no spin)
- ****Surface classification****: Identifies where the ball bounced (table, racket, floor, or other objects)

**** (3) CNN F1 Scores:****

- Spin: 0.96
- Surface: 0.97

**** (4) Baseline Numbers:****

	Method	Spin Precision	Spin Recall	Spin F1	Surface Precision	Surface Recall	Surface F1

```
|-----|---|---|---|---|
| GMM | 0.22 | 0.25 | 0.17 | 0.09 | 0.06 | 0.06 |
| SVM | 0.96 | 0.92 | 0.93 | 0.94 | 0.93 | 0.94 |
```

(5) Dataset Size:

- Total racket bounce samples: 3,396 (from 10 racket configurations)
- Additional non-bounce sounds: 1,239 (table, floor, other objects)
- 80/20 training/test split

(6) Performance Claims:

The paper states: "The CNN achieves the best performance over all metrics, followed closely by the SVM." It notes the GMM "performs extremely poorly" but does not use language like "strongly outperforms." The framing emphasizes practical applicability: "This pipeline may be integrated to table tennis robots for faster spin estimation."

5. assistant (2026-06-06T17:36:57.458Z)

The source confirms the numbers. Now let me check the key issues: (1) the domain is table tennis, not badminton, and (2) the claim says "racket-bounce sound" ? let me verify what's actually being classified and whether the "strongly outperforms" framing is supported. Let me check the SVM gap more carefully and search for the broader context.

6. user (2026-06-06T17:37:10.213Z)

Methodology and Limitations Analysis

(1) Classification Pipeline: Peak Detection ? Classification

Yes, the pipeline is **sequential and dependent**. Peak detection must occur first:

> "First, we use energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks."

The two-stage approach isolates candidate bounces via energy thresholding before the CNN refines classification. This design achieves millisecond-level temporal precision unavailable from frame-based classification alone.

(2) Recording Setup: Controlled Laboratory Conditions

The data collection was highly controlled:

> "The bounce sounds were recorded with the Zoom H4n Pro Handy Recorder, a directional microphone, at 44.1kHz ensuring consistent high-quality data. The microphone was always oriented towards the player at distances ranging from 50 cm to 2 m."

This represents **laboratory conditions without realistic match noise**?no crowd sounds, commentary, or competing acoustic environments. The authors explicitly note: "Due to the noise present in online recordings, we recorded our own dataset."

(3) Recording Personnel and Distribution Shift Risk

The paper does **not clearly specify** whether a single person performed all recordings or multiple players. The acknowledgments mention "Jonas Tebbe for sharing his table tennis experience and his help for the serve recordings," suggesting limited personnel involvement.

Critical concern: All samples derive from controlled, single-microphone conditions. Testing on the same distribution without cross-validation against real-world video would not demonstrate generalization to broadcast matches or varying acoustic environments.

(4) Scope: Spin Classification vs. Event Detection

The work addresses **both tasks but emphasizes spin classification**:

- Peak detection handles temporal localization ("millisecond-level temporal accuracy")
- Classification focuses on spin presence/racket type after peaks are isolated

The paper does not deeply explore broader event detection challenges beyond rejecting non-bounce sounds.

(5) Stated Limitations

The authors explicitly acknowledge:

> "One cannot judge the quality of a racket stroke from only the sound. Moreover, while the amount of spin can be determined, predicting the direction of the spin is considerably more difficult and requires visual input."

They also note that backspin and topspin are difficult to distinguish: "the backspin and topspin spectrograms, though different from no spin spectrogram, are almost identical."

7. assistant (2026-06-06T17:37:13.664Z)

Now I have a clear picture. Let me do a couple of searches to check for any independent corroboration or dispute, and to confirm whether this is a recognized credible result vs. an isolated, controlled-lab benchmark.

8. user (2026-06-06T17:37:21.799Z)

Web search results for query: "table tennis spin estimation bounce sound CNN Mel spectrogram arxiv 2409.11760"

Links: [{"title": "(PDF) Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://www.researchgate.net/publication/384115010_Spin_Detection_Using_Racket_Bounce_Sounds_in_Table_Tennis"}, {"title": "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline This work was funded by Sony AI", "url": "https://arxiv.org/html/2409.11760"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://arxiv.org/pdf/2409.11760v1"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.", "url": "https://arxiv.org/html/2409.11760v1"}, {"title": "[2409.11760] Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline", "url": "https://arxiv.org/abs/2409.11760"}, {"title": "[2409.11760v1] Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://arxiv.org/abs/2409.11760v1"}, {"title": "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline", "url": "https://arxiv.org/pdf/2409.11760"}]

Based on the search results, here's what I found about this research paper:

Paper Overview

The paper addresses spin detection in table tennis using sounds produced during racket-ball impacts, focusing on three key problems: precise bounce detection with millisecond-level temporal accuracy, classification of bounce surface (e.g., racket, table, floor), and spin detection from audio alone.

Methodology

The research proposes a real-time-capable pipeline that combines energy-based peak detection with convolutional neural networks trained on a novel dataset of 3,396 bounce samples recorded across 10 racket configurations.

Detected segments are converted to Mel spectrograms and classified using CNNs to identify the bounce surface and detect whether spin was applied.

Technical Details

A six-layer CNN was utilized for bounce sound classification, with each layer consisting of a 2D convolution, ReLU activation, and batch normalization. Experimental results demonstrated that the CNN performed best using 64 Mel bands and an FFT window length of 256 samples.

Results

The system achieves millisecond-level onset accuracy. The system achieves accurate and low-latency detection of bounces, and reliably classifies both the surface of contact and whether spin was applied.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-06-06T17:37:21.912Z)

Web search results for query: "audio shot hit detection badminton broadcast video noise crowd robustness accuracy"

Links: [{"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "Shot detection using skeleton position in badminton videos - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2021SPIE11766E..1KY/abstract"}, {"title": "(PDF) Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://www.researchgate.net/publication/322075902_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "An AI-based badminton smash detection using residual-shuffle network optimized based on upgraded pufferfish optimizer - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/pii/S2090447925001558"}, {"title": "[2307.16000] Automated Hit-frame Detection for Badminton Match Analysis", "url": "https://arxiv.org/abs/2307.16000"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "Wearable Audio and IMU Based Shot Detection in Racquet Sports", "url": "https://arxiv.org/pdf/1805.05456"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}]

I've completed the web search for your query about audio shot hit detection in badminton broadcast videos with attention to noise and crowd robustness. Here are the key findings:

Audio-Based Hit Detection for Badminton

Research employs lightweight, audio-based detection models that train neural networks to identify the distinct acoustic signatures of racquet-ball impacts from broadcast audio tracks.

Robustness Against Ambient Noise

A critical challenge in broadcast environments is handling crowd noise. To ensure models are robust against complex ambient stadium noises, training data is augmented by overlaying various background sounds (e.g., crowd cheering, clapping, and announcer voices) onto audio samples, allowing the model to reliably distinguish true hitting events from similar impulsive noises in the broadcast.

Video-Based Approaches

Beyond audio-only methods, video-based hit detection networks using court, pose, and 2D shuttlecock information can reach 86.3% accuracy. Additionally, research has achieved over 92% accuracy for applying player keypoints sequences on shuttlecock detection and 99% accuracy on shot angle recognition for badminton match analysis.

Multimodal Approaches

Acoustic cues such as ball strikes often align with event boundaries and provide contextual signals that may not be easily inferred from visual frames alone, and fusing modalities aims to

improve robustness and temporal precision.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-06-06T17:37:39.467Z)

I now have all the evidence I need. Let me assess the claim against the verification checklist.

Key findings:

1. ****Numbers are accurate****: The CNN F1 scores (spin 0.96, surface 0.97), SVM (0.93?0.94), and GMM (0.06?0.17) all match the source exactly. The "six-layer CNN", "64 Mel bands", "FFT window 256" details are also verbatim correct.
2. ****Domain mismatch (minor)****: The paper is about ****table tennis****, not badminton. The research question is about badminton/racket sports broadly, so "racket-bounce sound events" is technically defensible, but the claim presents it as broadly applicable evidence.
3. ****"Strongly outperforming" SVM is an OVERREACH****: The paper itself says the CNN is "followed closely by the SVM." SVM achieves 0.93?0.94 F1 vs CNN's 0.96?0.97 ? a 2?3 point gap. That is NOT "strongly outperforming." The dramatic gap is only versus GMM (0.06?0.17). The claim conflates "strongly outperforms GMM" with "strongly outperforms classical methods (SVM ~0.93?0.94)" ? but SVM is a classical method and is nearly competitive. This is a misread.
4. ****"Simple locally-runnable models suffice" ? context-stripped****: The 0.96/0.97 numbers come from a ****controlled lab dataset**** (Zoom H4n Pro directional mic, 50cm?2m, "Due to the noise present in online recordings, we recorded our own dataset"). No crowd noise, no broadcast conditions. For a self-hosted pipeline processing real match video (the actual context), these numbers do NOT transfer. The paper explicitly avoided noisy real-world audio.
5. ****Different task than rally segmentation****: The paper classifies ****spin/surface**** of an already-isolated bounce ? not detecting WHEN hits occur in noisy video, which is the relevant task for rally segmentation. The claim's framing ("once peaks are isolated") acknowledges this but then over-generalizes to "audio event classification."

The claim is refuted on multiple grounds: (a) "strongly outperforming SVM" contradicts the paper's own "followed closely by the SVM" + the near-identical numbers; (b) "simple models suffice" generalizes a controlled-lab spin-classification result to real-world audio event classification, which the source explicitly does not support (they avoided noisy data).

11. user (2026-06-06T17:37:46.943Z)

Structured output provided successfully

12. assistant (2026-06-06T17:37:54.404Z)

I've completed the adversarial verification and returned my structured output.

****Verdict: refuted = true**** (high confidence)

The claim's numbers are accurate, but it contains two material overreaches plus a domain/task mismatch:

1. ****"Strongly outperforming SVM"***** contradicts the paper's own words ? it says the CNN was "followed closely by the SVM" (0.96 vs 0.93?0.94, a 2?3 point gap). The dramatic outperformance is only versus GMM. SVM is itself a classical method and is nearly on par, so "simple models strongly beat classical methods" is a misread.
2. ****"Simple models suffice for audio event classification"***** is context-stripped ? the 0.96/0.97 figures come from a controlled lab dataset (directional Zoom mic at 50cm?2m). The authors explicitly recorded their own data "due to the noise present in online recordings," i.e., they deliberately avoided real-world noisy audio. Those numbers do not transfer to broadcast/match-video conditions.

3. ****Domain/task mismatch**** ? it's table tennis (not badminton), and the task is spin/surface classification of an already-isolated peak, not detecting when hits occur in noisy video (the rally-segmentation-relevant problem).

The paper is a legitimate, credible Sony AI source and supports the weaker directional point (CNN-on-Mel is locally feasible), but not the strong "strongly outperforms / suffices" framing as stated.